A Comparative Analysis of Hyperparameter Sensitivity Within the MNIST Dataset Across Six Classification Models

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Abstract

This study examines how hyperparameter search variability affects classifier performance evaluation, by analysing six standard algorithms (logistic regression, KNN, decision tree, random forest, linear SVM, and naïve Bayes) on a subsampled MNIST dataset (LeCun et al., 1998). Using randomised hyperparameter search, each model is trained under 30 random configurations and evaluated by 3-fold cross-validation. Unlike conventional reporting of only the best test accuracy, we focus on the full distributions of accuracies. The results show substantial variance, such as a range between 86.6-93.1% (mean accuracy of 91.1%) for random forests and a range between 89.2-92.8% (mean accuracy of 91.0%) for KNN, which echo concerns found in the literature that some single "best" accuracies are often misleading (Jordan, 2024; Slawski et al., 2008), and recommend reporting mean and confidence intervals over multiple runs and evaluating stability to improve reproducibility and provide a more robust and stable depiction of model reliability. We highlight that a model with lower peak performance but higher consistency may be preferable, and that performance evaluation should account for variability to avoid excessively optimistic conclusions.

1. Introduction

Machine learning research heavily relies on standardised datasets to evaluate an algorithm's performance. The MNIST dataset, a series of $70,000$ greyscale 28×28 pixel images of handwritten digits (LeCun et al., 1998), is one of the most common datasets for image classification (Cohen et al., 2017). Many machine learning models like CNNs can easily achieve high scores accuracy on the dataset (Cireşan et al., 2011). In order to better improve these models, the user has to rely on tuning hyperparameters like learning rate, batch size, number of layers, etc.

However, this raises an issue in machine learning where we evaluate based on these tuned results. It has become a common practice to perform extensive grid or randomised searches across a vast hyperparameter space, only to publish the results with the single highest test accuracy achieved (Bergstra & Bengio, 2012; Saidi et al., 2025). By reporting only peak performance, the results end up becoming obscured and potentially meaningless (Cawley & Talbot, 2010). For example, suppose an algorithm that hits a peak accuracy of 97% on a "lucky run" but averages 85% across most configurations would be inferior to a model that consistently yields 95%. However, due to leaving out the less than ideal results, we may be led to believe in misleading statistics. Standard reporting metrics fail to capture let alone note this critical distinction, contributing to broader issues of reproducibility in the field.

This project explores the reliability of accuracy reporting by testing the performance variance of six distinct traditional machine learning algorithms: Logistic Regression, K-Nearest Neighbours, Decision Trees, Random Forests (Breiman, 2001), Linear Support Vector Machines (Cortes & Vapnik, 1995), and Naïve Bayes. By simulating randomised hyperparameter searches on the MNIST dataset, while also taking note of the complete distribution, this study aims to demonstrate that a single maximum score is insufficient and potentially deceptive metric for algorithmic evaluation. Ultimately, this report illustrates the necessity of distribution-based evaluation to ensure true model robustness.

2. Methodology

The main objective of the experiment was to explore randomized hyperparameters and record the results. In order to achieve this, we applied a randomised search framework across a set of classification algorithms.

2.1. Dataset and Preprocessing

The models were evaluated using the MNIST database of handwritten digits. The MNIST database consists of 28×28 greyscale images of handwritten digits (LeCun et al., 1998). We further flattened the two-dimensional spatial structure into 784-dimensional feature vectors.

In order to ensure stability and convergence, the pixel values were normalised from $[0, 255]$ into a continuous range from $[0.0, 1.0]$. This is needed especially for models like Support Vector Machines and Logistic Regression. In order to make the randomised hyperparameters feasible, we subsampled 5000 training instances.

2.2. Algorithm Selection

Choosing multiple algorithms allows us to explore a broad spectrum of classification paradigms. This allows us to see how hyperparameter variance can impact results. We chose Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest (Breiman, 2001), Linear Support Vector Machine (SVM) (Cortes & Vapnik, 1995), and Gaussian Naïve Bayes.

2.3. Experimental Design (Randomised Search)

Our experiment heavily relies on the `RandomizedSearchCV` class from the `sklearn.model_selection` library (Scikit-Learn Developers, 2026). This library randomly samples combinations of hyperparameters from given ranges. These hyperparameters are then trained and evaluated using cross-validation (set to 3 here) and finds the best performing combination. For each of the six algorithms, there are 30 distinct hyperparameter configurations with train-test splits applied as well. This resulted in 90 total fits per model resulting in a significant sample size to map the accuracy distribution.

2.4. Evaluation Strategy

Unlike most machine learning research where the evaluation phase focuses on the highest-performing score, we are instead taking into account all scores. We will be recording the mean accuracy of all 30 runs per model then plotting these on a graph. This allows us to explore the robustness of the hyperparameter optimisation.

2.5. Hypotheses

2.5.1. Claim 1: Variance

Performance across randomly sampled hyperparameters exhibits substantial variance within each model class.

2.5.2. Claim 2: Ranking instability

Model rankings based on maximum accuracy differs from rankings based on mean accuracy.

2.5.3. Claim 3: Sensitivity

Model rankings differ significantly in hyperparameter sensitivity.

3. Results

After running our code (available at https://github.com/yassataiseer/AMATH445-Final-Project-Paper) with the randomised hyperparameters search, we ended up with results showcasing the cross-validation accuracy, the variance, as well as min and max. We end up with the following results:

Machine learning model	Minimum accuracy	Mean accuracy	Maximum accuracy	Variance spread
Logistic Regression	85.64%	88.81%	89.96%	4.32%
K-Nearest Neighbours	89.24%	91.04%	92.78%	3.54%
Decision Tree	67.38%	73.49%	75.62%	8.24%
Random Forest	86.62%	91.09%	93.06%	6.44%
Linear SVM	84.28%	86.82%	89.58%	5.30%
Naïve Bayes	57.92%	59.97%	62.92%	5.00%

Table 1: The results as reported by the code.

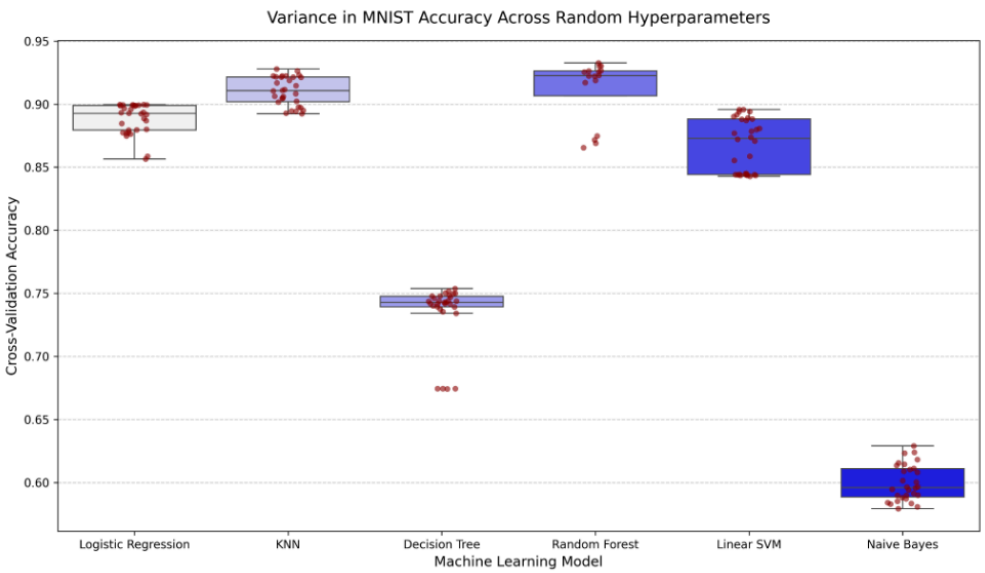

Figure 1: Variance in MNIST accuracy across hyperparameters

The box plot illustrates the interquartile range and median for each algorithm. Alongside, overlaid scatter points that represent the individual accuracy scores for all 30 randomly sampled configurations.

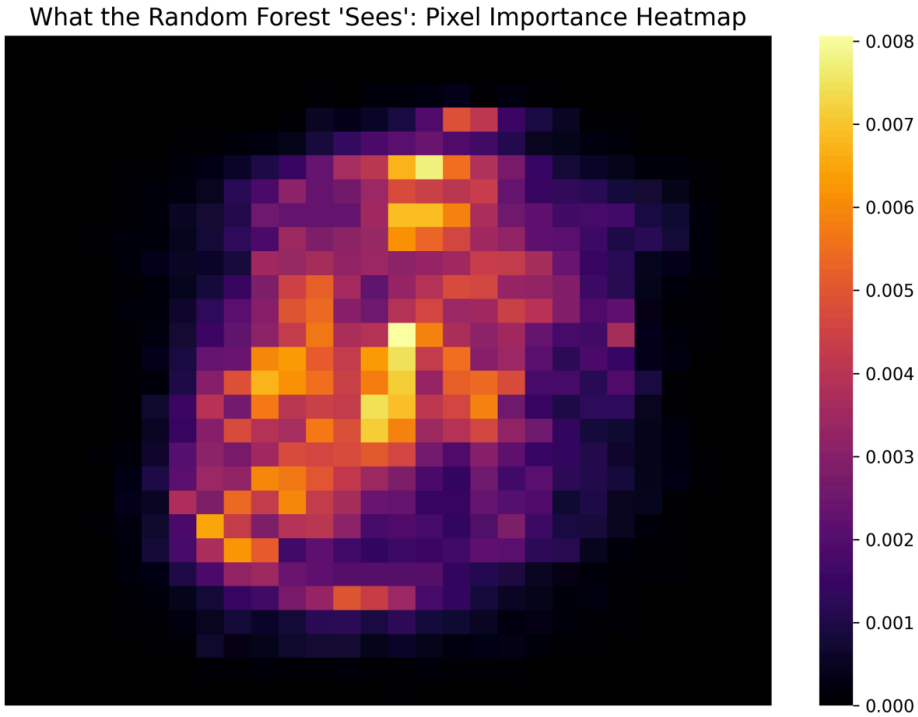

Figure 2: Pixel importance heat map

When training the Random Forest, we calculate a "feature importance" score for every single variable. This image maps those 784 scores back into a 28×28 grid.

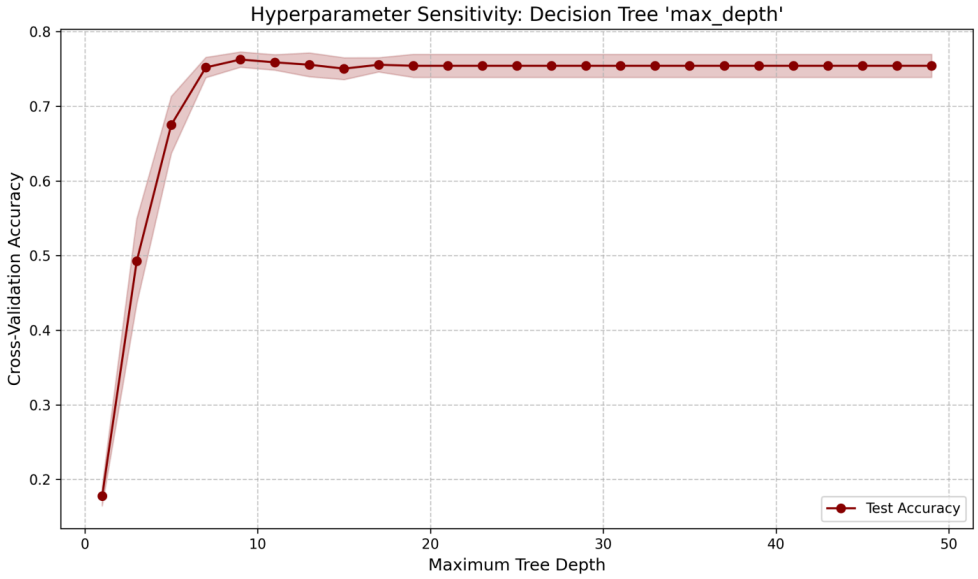

Figure 3: Hyperparameter sensitivity of the decision tree

This line graph tracks what happens to the Decision Tree's accuracy as we increase the maximum allowed depth of the tree. The shaded red area represents the variance (standard deviation) across the cross-validation folds.

4. Analysis

The objective of this study was to show how accuracy on the MNIST dataset can variate based on hyperparameter selection. The box plot shows us significant vulnerabilities in relying on optimised test scores for algorithmic comparison. The data demonstrates a model's accuracy is often an outlier, for example in the Random Forest ensemble achieved the highest performance at 93.06% however, its mean was at 91.09%. This 2% gap is non-trivial and would be ignored in most research cases.

The variance spread in the table (Table 1) shows the algorithm's reliance on hyperparameter tuning. For example, the Decision Tree shows very high variance at 8.24% and a 67.38% accuracy in the suboptimal conditions. This is due to shallow trees underfitting the data of a handwritten digit; this high sensitivity shows how its integrity is based on aggressive tuning. On the other hand we notice that KNN had a much smaller variance spread at 3.54% showing that distance based spatial classification remains highly robust regardless of specific hyperparameters. As seen in Table 1 and Figure 1, models exhibit non-trivial spreads between minimum and maximum accuracies, indicating that performance is not concentrated around a single value, but rather distributed across a range of outcomes, supporting our first claim.

An overlap can also be noticed between the accuracy and the different algorithms. As seen in the box plots, the absolute lowest accuracy recorded for the Random Forest (86.62%) underperformed the mean accuracy of the standard Logistic Regression model (88.81%). Furthermore, the interquartile range of the Linear SVM overlaps almost entirely with the lower quartile of the Random Forest. This further shows that comparing a heavily optimised algorithm to an unoptimised baseline is misleading.

Despite the Random Forest performing best in maximum accuracy, KNN slightly outperforms Random Forest while Logistic Regression remains competitive relative to models with higher peak scores, while overlap in the distributions (Figure 1) shows that lower-performing configurations of higher-ranked models often fall below the average performance of more stable models (for example, the minimum accuracy of Random Forest is lower than that of the mean average of Logistic Regression). This supports our second claim, that model rankings are not invariant and depend strongly on whether peak or average performance is used.

The visual evidence found from the heat map (Figure 2) and line graph (Figure 3), further exposes the structural limitations of applying traditional algorithms to spatial data. The hyperparameter sensitivity curve for the Decision Tree, mathematically illustrates why non-convolutional models exhibit such extreme variance. With shallow depth (`max_depth < 5`), the tree severely underfits the spatial boundaries required to separate digits, resulting in sub-70% accuracy. Even when heavily tuned, these models suffer from a fundamental lack of spatial awareness. The 28×28 images must be flattened into 784-dimensional 1D vectors, as the models cannot recognize contiguous lines or loops. As shown in the Random Forest feature importance heat map (Figure 2), the ensemble model simply assigns high probability weights to the absolute spatial coordinates in the center of the grid.

The variance spread reported in Table 1 shows the largest spread in Decision Trees (with 8.24%), indicating a high sensitivity to hyperparameter choices, such as maximum depth, while KNN exhibits a smaller spread (3.54%), suggesting greater resilience to parameter variation,

with Random Forest and Linear SVM somewhere between these extremes. This supports our third claim, that models differ significantly in hyperparameter sensitivity.

Overall, our results consistently demonstrate that hyperparameter variability plays a critical role in model evaluation. Across our three claims, performance is shown not only to be model-dependent but also configuration-dependent. This highlights the particular importance of distribution-based reporting.

5. Suggested Future work

Given the limited scope of the paper, there are tremendous pieces left wanting of additional research with plenty of open avenues for approach. The limitations in our work include the smaller sample size and working only in a single dataset. As noted by Saidi *et al.*, reported accuracy often inversely correlates with sample size, and a larger training set and multiple metrics could potentially influence conclusions (Saidi et al., 2025). We list suggestions for such potential work.

We may wish to repeat this analysis on additional benchmarks, such as *Fashion-MNIST* (Xiao et al., 2017) or another tabular dataset, which may demonstrate whether observed patterns hold beyond those demonstrated in the MNIST set. It was shown with CIFAR-10 that in general, fresh test sets may differ from those of the original findings (Recht et al., 2018), and similarly replicating the experiments on new data may reveal additional insight on reliability of our algorithms.

Using a feed-forward neural network model, such as a 2-layer MLP or a simple CNN may provide additional insight, due to the high sensitivity of deep models to initialisation and hyperparameters (Jordan, 2024). An investigation of the accuracy distributions would supplement our understanding more than what was provided using just classical algorithms.

It is also further desirable to investigate using confidence intervals and statistical tests, such as a paired t-test or Wilcoxon test across cross-validation folds to see if differences in mean accuracy are significant. This would address our limitations of unquantified uncertainty.

6. Conclusion

In this project, we investigated the validity of single point accuracy reporting by evaluating the hyperparameter variance of different machine learning models on the MNIST dataset. Through trials of randomised hyperparameters, we noticed that peak performance often represented outliers rather than reliable indicators of the mean. Even though methods like Random Forest achieved the highest absolute accuracy, their significant variance spread reveals underlying stability problems which are only ignored when only “lucky runs” are published.

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